

## CALCULUS $\mathcal{I}$ – CP 1 REVISION SHEET FOR THE FINAL EXAM

### Exercise 1.

1. Without solving the following equation, show that it admits a unique solution  $\alpha$  in the interval  $] \frac{3}{2}, 2[$

$$x = \frac{1}{2} \left( x + \frac{3}{x} \right).$$

2. Let  $(u_n)_{n \in \mathbb{N}}$  be the real sequence defined by :

$$\begin{cases} u_0 = 2, \\ u_{n+1} = \frac{1}{2} \left( u_n + \frac{3}{u_n} \right). \end{cases}$$

- (a) Show that, for all  $n \in \mathbb{N}$ ,  $u_n - \sqrt{3} > 0$ .
- (b) Show that the sequence  $(u_n)_{n \in \mathbb{N}}$  is decreasing.
- (c) Deduce that the sequence  $(u_n)_{n \in \mathbb{N}}$  is convergent.
- (d) What does  $\alpha$  represent for the sequence  $(u_n)_{n \in \mathbb{N}}$ ? Give its value.
- (e) Let  $S = \{u_n \mid n \in \mathbb{N}\}$ . Determine the supremum, the infimum, the maximum, and the minimum of  $S$ , if they exist.

**Exercise 2.** Let  $a, b$  be two real numbers and let  $f$  be the function defined by

$$f(x) = \begin{cases} x^2 + b, & \text{if } x \geq 0, \\ \frac{\sin(ax)}{x}, & \text{if } x < 0. \end{cases}$$

1. Give the relation between  $a$  and  $b$  such that the function  $f$  is continuous and differentiable on  $\mathbb{R}$ .
2. Compute the derivative of  $f$ .
3. Is the function  $f$  of class  $C^1$  on  $\mathbb{R}$ ?

**Exercise 3.** Let  $g$  be the function defined by

$$g(x) = \begin{cases} 0 & \text{if } x = 0, \\ x + \frac{x \ln(x)}{1-x} & \text{if } 0 < x < 1, \\ 0 & \text{if } x = 1. \end{cases}$$

1. Show that  $g$  is continuous on  $[0, 1]$ .
2. Show that there exists a point  $c \in [0, 1]$  such that  $g'(c) = 0$ .

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**Exercise 4.**

1. Compute the following limits :

$$\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x-3}), \quad \lim_{x \rightarrow 0^+} \frac{x^2 + 2|x|}{x}, \quad \lim_{x \rightarrow +\infty} \frac{1}{x} E(x).$$

2. Let  $f$  be the function defined by :

$$f(x) = x^2 \cos\left(\frac{1}{x}\right), \quad x \in \mathbb{R}^*.$$

Show that  $f$  is extendable by continuity at point 0.